

Class Test II
CITS2200 Data Structures and Algorithms
May 9, 2024

- Total marks for this paper is 10. Please answer by filling the appropriate circle for each question in your answer sheet.
- You should carefully fill your student number in the answer sheet.
- Please return only the answer sheet, do not return the question paper. Take it away with you and throw it in a recycling bin.

Q1. A cyclist can travel 24 km/hour when wind is blowing from behind, and 12 km/hour against the wind. The cyclist travels *A* to *B* and then *B* to *A*. The wind is blowing from *A* to *B* throughout the travel of the cyclist. What is the amortized speed of the cyclist? (b)

(a) 12 km/hour; (b) 16 km/hour; (c) 20 km/hour; (d) Impossible to say without knowing the distance from *A* to *B*.

Q2. Each node of a doubly-linked list can store an item and two references *next* and *previous* to the nodes to the right and left of the node respectively. There are three nodes *A*, *B* and *C* in a doubly-linked list, such that *A.next*=*B* and *B.next*=*C*; *C.previous*=*B* and *B.previous*=*A*. We have used the names of the nodes as their references. We want to delete the node with reference *B*. Which one of the following is a correct way to delete *B* from the doubly-linked list?

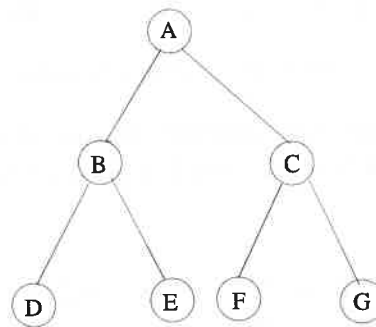
- (a) *A.next*=*B.next*
 C.previous=*A.next*
- (b) *C.previous.previous.next*=*C*
 C.previous=*C.previous.previous*
- (c) *C.previous.next*=*C*
 C.previous=*C.previous.previous*
- (d) *A.next*=*B.next*
 C.previous=*B.previous.next*
- (b)

Q3. Each node of a singly-linked list can store an item and a reference *next* to the node to its right. There are two nodes *A* and *C* in a linked-list such that *A.next*=*C*. You have created another node whose reference is *B*. Which of the following connects *B* correctly between *A* and *C*?

- (a) A.next=B (b) A.next=C (c) A.next=B (d) B.next=C
 B.Next=C B.next=C B.next=A A.next=C

(a)

Q4. Which is the correct **inorder** enumeration of the nodes of the following binary tree if you start traversal from the root node A



(d)

- (a) A - B - C - D - E - F - G; (b) A - B - D - C - E - F - G; (c) B - A - D - E - C - F - G; (d) D - B - E - A - F - C - G.

Q5. You have an input array storing the items A, B, C, D, E from index 0 to index 4. You first enqueue these items from index 0 to index 4. Now you dequeue the items one by one and push them into a stack. Finally, you pop the elements from the stack one by one. The order of the popped items is:

(d)

- (a) E, D, C, A, B; (b) A, B, C, D, E; (c) B, A, C, D, E; (d) none of the above.

Q6. You are storing integers 26, 17, 14, 8, 5 in an array of size 5, using a hash function $N \bmod 5$ when N is the integer you are storing. Which of the following is a correct representation of the array after storing these integers?

- (a)

17	26	14	8	5
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 (b)

26	17	14	8	5
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 (c)

5	26	17	8	14
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 (d)

5	17	14	8	26
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(c)

Q7. A **max-heap** is stored in an array as a binary tree, with the highest priority item at the root. The array location with index 0 is empty, and all other array locations store the priorities of the elements in the **max-heap** in the usual way of storing a binary tree using an array. Which of the following arrays correctly represents a **max-heap**?

(a)		45	13	24			34	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(b)		45	13	7			24	34					13	3						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(c)		45	13	34			24	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(d)		45	24	13			34	21					7	3						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(c)

Q8. The highest priority item is now deleted from the **max-heap** and an item with priority 16 is inserted. Which of the following correctly represents the new heap after rearrangement?

(a)		45	13	24			34	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(b)		16	13	34			24	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(c)		34		13	24		16	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(d)		34	13	24			16	21					3	7						
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19

(b)

Q9. Which of the following statement is false for a singly-linked list? Assume that the **next** references go from left to right, and you are given a reference called **current** of a node in the middle of the linked list and there are n nodes to the left and right of **current**. You are also given the reference of the **first** node of the linked list.

(c)

- (a) The complexity of finding the reference of a node 5 nodes to the right of **current** is $O(1)$;
- (b) The complexity of adding a node to the right of **current** is $O(1)$;
- (c) The complexity of adding a node to the left of **current** is $O(1)$;
- (d) The complexity of counting the number of nodes in the linked list is $O(n)$.

Q10. Which of the following statement is true for a doubly-linked list? Assume that you are given a reference called **current** of a node in the middle of the linked list and there are n nodes to the left and right of **current**. You are also given the reference of the **first** node of the linked list.

- (a) The complexity of counting the number of nodes in the doubly-linked

list is $O(\log n)$;

(b) The complexity of adding a node to the left or right of **current** is $O(1)$;

(c) The complexity of counting the number of nodes in the doubly-linked list is $O(1)$. (b)

(d) The reference of the last node in the doubly-linked list can be found in $O(1)$ time.

End of Paper
